MOHAMED AMINE TRABELSI

Data & AI Engineering
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PROFILE SUMMARY

Data & AI Engineering student (Bac+5 / Master-level Cycle d'Ingénieur) with hands-on experience building full-stack applications, Retrieval-Augmented Generation (RAG) pipelines, and computer vision systems across four industry internships. Skilled at architecting provider-agnostic LLM integrations (Mistral, Gemini, MedGemma) and delivering production-ready APIs with FastAPI, Django, and Flask. Solid foundation in software engineering, data engineering, and DevOps, with a proven ability to design end-to-end pipelines from data ingestion through deployment, backed by 5 industry-recognized DataCamp certifications. Seeking a challenging internship to apply this expertise to impactful, scalable AI-driven solutions.

PROFESSIONAL EXPERIENCE

06/2026 - 08/2026 | IdeaLab — Bizerte, Tunisia
AI / RAG Engineer (Mistral / Gemini) & Full-Stack Developer (Python / TypeScript)

- Engineered Digital Campus, a full-stack academic platform, implementing JWT authentication with email verification, student profile/onboarding management, and a REST API for chat sessions and conversation history.
- Architected a provider-agnostic RAG module spanning 2 LLM providers (Mistral primary, Google Gemini fallback) that answers exclusively from verified documents with precise citations, leveraging PostgreSQL and pgvector for semantic search.
- Delivered the backend in Python (FastAPI) and SQLAlchemy and the frontend in Next.js/TypeScript, and automated the test suite with pytest.
  Stack: *Python, FastAPI, SQLAlchemy, PostgreSQL, pgvector, Mistral, Gemini, Next.js, TypeScript, pytest*

06/2026 - 08/2026 | TechCodex — Ariana, Tunisia
AI Engineer - Computer Vision & Full-Stack Developer

- Fine-tuned and 4-bit quantized a 4-billion-parameter Vision-Language model (MedGemma 1.5 4B), cutting memory footprint while extracting structured data from handwritten and printed medical prescriptions with optimized local GPU inference.
- Built a full-stack architecture connecting a Flask AI backend, a Django REST Framework business API, and a React/Vite frontend for reviewing and validating extraction results.
- Implemented a human-in-the-loop validation workflow to guarantee data reliability before export.
- Automated PDF export of validated prescriptions and the full data lifecycle, including anonymization, archiving, and purging.
  Stack: *Python, MedGemma, PyTorch, Flask, Django REST Framework, React, Vite, Docker, GPU (CUDA)*

06/2025 - 08/2025 | LabelGarantie — Bizerte, Tunisia
AI Engineer - NLP & Computer Vision

- Trained automatic text-extraction models to process complex documents (PDFs, images) by combining OCR with LLM architectures.
- Designed and integrated an end-to-end pipeline covering ingestion, parsing, classification, and Retrieval-Augmented Generation (RAG).
- Industrialized the complete pipeline by exposing it through a REST API built with FastAPI.
  Stack: *Python, PyTorch, HuggingFace, LangChain, Tesseract, PaddleOCR, FastAPI, REST API*

06/2024 - 08/2024 | LabelGarantie — Bizerte, Tunisia
Front-End Web Developer

- Integrated dynamic, responsive UI modules that improved the overall user experience.
- Applied the MVC pattern in Symfony and connected the front end to REST APIs for seamless data exchange.
  Stack: *Symfony (PHP), Twig, JavaScript, HTML5, CSS3, MySQL, Git*

PERSONAL PROJECTS

Portail Étudiant Intelligent (FSB) | *Jan 2026 - May 2026*
Full-stack university portal with secure grade management, timetables, signed PDF documents, and a news feed. Integrated a Gemini AI assistant for academic guidance and personalized performance analysis. Built with TypeScript, React, Vite, Tailwind CSS, and Supabase; deployed on Vercel.
Link: github.com/AymenJeddou/FSBridge

JobJet - Professional Social Network Platform | *Dec 2025 - Jan 2026*
Full-stack professional networking application with profile management, job postings, applications, real-time messaging, and notifications, plus automatic PDF CV generation and admin/moderation modules. Built with Node.js, Express.js, MongoDB, and Docker.
Link: github.com/AymenJeddou/JobJet

DevOps CI/CD Pipeline & Infrastructure Automation | *Dec 2025 - Jan 2026*
End-to-end CI/CD automation with GitHub Actions and Docker containerization, configuration management with Terraform and Ansible, and post-production monitoring with Prometheus and Grafana.
Link: github.com/AbderrahimNeji/FinalProjectDevops

OliveGrove IoT - Smart Farm Monitoring | *Sep 2025 - Nov 2025*
End-to-end IoT solution for real-time monitoring of environmental parameters (temperature, humidity) in an olive grove, with automated sensor data collection and a web-based visualization dashboard.
Link: github.com/AbderrahimNeji/OliveGrove_IOT

TECHNICAL SKILLS

AI & Data: RAG pipelines, LLM integration (Mistral, Gemini, MedGemma), Computer Vision, NLP
Backend & Web: Python, FastAPI, Django REST Framework, TypeScript, React/Next.js
Data & DevOps: PostgreSQL (pgvector), MongoDB, Docker, Git

EDUCATION

09/2024 - Present | Université de Carthage — Engineering Degree, Data Specialty (Cycle d'Ingénieur)
09/2022 - 05/2024 | Université de Carthage — Integrated Preparatory Cycle - Mathematics, Physics, Computer Science

CERTIFICATIONS

- Associate AI Engineer for Data Scientists - DataCamp (Project-Based) — View certificate
- Associate AI Engineer for Developers - DataCamp (Work-Based) — View certificate
- Associate Data Scientist - DataCamp (Project-Based) — View certificate
- Associate Data Engineer - DataCamp (Project-Based) — View certificate
- Associate Data Analyst - DataCamp (Project-Based) — View certificate

LANGUAGES

Arabic: Native · French: Professional Proficiency (B2) · English: Professional Proficiency (B2)